Pranshu Tripathi

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COVER LETTER

Dear Admissions Committee,

I am writing to express my strong interest in pursuing a Master's degree in Computer Science at Georgia Institute of Technology. With an in-depth understanding of deep learning gained through the coursework of Masters in Computer Science, I am eager to contribute in applying these principles in scientific software development.

At Schrödinger, I've had the opportunity to work on core product features, backend infrastructure, and AI-powered tools. I recently led backend development for a voice-enabled LLM assistant, built using FastAPI, GCloud Run, and Docker, now deployed internally across teams. Previously, I improved core molecular interaction speeds by hundreds of times through a modular C++ library, streamlining operations that once took minutes to just seconds. These projects have strengthened my belief in the value of well-structured, maintainable systems.

My graduation from Indian Institute of Information Technology, Allahabad, with a B.Tech in Electronics and Communication Engineering has laid the foundation for core Computer Science concepts. I have consistently excelled in my coursework, maintaining a strong academic record. In my final year at IIIT Allahabad, I did research projects under supervision of Prof. Dr. Shanti Chandra that were focused on supervised learning on CNN and LSTM models, with signal processing on the audio form datasets. I have all the necessary prerequisites for this program, including Data Structures, Algorithms, Operating Systems and Computer Networks. Hence I believe that I am well-equipped to handle the rigour of the curriculum at Georgia Institute of Technology.

I have recently got the opportunity to work on the Large Language Model based agents at Schrödinger, which has further fueled my interest in deep learning and its applications. The renowned coursework at Georgia Institute of Technology, will particularly help understand the complex internal functioning of these architectures which are currently a black box to me.

I am keen on joining the vibrant academic community at Georgia Institute of Technology, take pride in building useful tools and contributing to cutting-edge research in Computer Science.

Thank you for considering my application.

Warm regards,
Pranshu Tripathi